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Evaluating green financing mechanisms for natural resource management: Implications for achieving sustainable development goals

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ABSTRACT

China has made noteworthy strides in achieving the Sustainable Development Goals (SDGs), including natural resource management. The country's economic expansion has put more strain on its natural resources during Covid-19 pandemic, which has contributed to environmental deterioration, the extinction of species, and climate change. Consequently, green finance can create new opportunities for assisting natural resource management practices that support environmental, social, and economic sustainability. In order to realize the SDGs, this study employs a hybrid Fuzzy Multi-Criteria Decision Making (MCDM) approach based on the Analytical Hierarchy Process (AHP) and Visekriterijumska Optimizacija i Kompromisno Resenje (VIKOR) to assess and prioritize important criteria, sub-criteria, and green financing mechanisms for natural resource management in China. Therefore, this study identified six criteria, eighteen sub-criteria, and six green financing strategies for promoting sustainable development and natural resource management methods in China. According to the Fuzzy AHP findings, financial sustainability (0.192), institutional feasibility (0.176), and environmental impact (0.171) are the most crucial criteria. While, the Fuzzy VIKOR findings reveal that carbon markets, green bonds, and green investment funds are the best financial solutions for reaching sustainable development and effective resource management practices in China. This research enables policymakers, investors, and other stakeholders improve sustainable development and natural resource management in China.

1. Introduction

The management of natural resources is essential for China's to achieve the Sustainable Development Goals (SDGs) (Li et al., 2022a). The country has a variety of habitats, including marshes, grasslands, forests, and oceanic regions. However, the burden on natural resources has intensified as a result of rapid economic growth, pollution, land degradation, biodiversity loss, and climate change (Chien et al., 2023). Furthermore, managing natural resources is a crucial for sustainable development, particularly in a country like China that faces great challenges as a result of its booming population, expanding urbanization, and industrialization. The country has been under tremendous pressure to meet the needs of its citizens while preventing the depletion of the natural resources because it has the largest population in the world (Shaheen et al., 2022). The management of natural resources entails making sure they are sustainable and conserved as well as improving how they are used for social and economic advancement. As a result, green financing strategies are required for China to achieve the SDGs. The SDGs are a set of 17 objectives that were established by the UN in 2015 with the intention of eradicating poverty, preserving the environment, and fostering prosperity for all (United Nations General Assembly, 2015). The most urgent issues facing the world now include climate change, poverty, inequality, and environmental degradation (Ali et al., 2022). The SDGs provide an integrated and comprehensive framework for cooperation between governments, corporations, and civil society to achieve a more sustainable future.

China has encountered new challenges in meeting SDGs and sustainably managing its natural resources in the context of Covid-19 and conflict. Worldwide, especially in China, the epidemic has placed enormous strain on economies and communities, causing disruptions in a number of different industries and escalating pre-existing environmental and social problems. Prior to the pandemic, China's economy was expanding, which may have put more strain on its natural resources and accelerated environmental deterioration, species extinction, and

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climate change (Cao and Solangi, 2023; Li et al., 2023). The Covid-19 epidemic, as well as any geopolitical tensions, could have complicated efforts to address these issues. The country's capacity to decide which sustainable development measures to prioritize and carry out may have been impacted by the focus and resources that were diverted to addressing the pandemic's health and economic effects as well as any potential interruptions in supply chains and trade.

In recent years, China has made significant progresses toward attaining the SDGs (Xu et al., 2023). The country has invested extensively in renewable energy and other sustainable development initiatives, and it has had exceptional economic growth that has helped millions of people escape poverty (Li et al., 2022b). China still has some huge efforts to do to manage its natural resources and protect the environment. For instance, the nation has struggled with air pollution, water scarcity, soil erosion, and biodiversity loss. Green finance mechanisms can be extremely important in efforts to manage natural resources in a way that promotes sustainable development. Green financing can offer additional resources, capacity, and knowledge to support initiatives to manage natural resources, which can aid in achieving sustainable development objectives. These financing options required to fund initiatives that encourage sustainable development, advance environmental protection, and stimulate the economy (Liu et al., 2020).

The lack of awareness and comprehension among important stakeholders, including government, investors, and the general public, is one of the major obstacles to the implementation of green financing mechanisms (Wasan et al., 2021). It is essential to enhance awareness of the potential benefits of various financial sources and to support capacity building for successfully utilization. Even though, the country has made significant development toward establishing laws and policies that encourage sustainable growth, more has to be done to encourage the establishment of green finance practices. This requires developing legal frameworks, boosting institutional capacity, and promoting transparency and accountability. Furthermore, broader policy frameworks and SDGs must include green finance structures (Lee, 2020). This needs coordination and collaboration among multiple stakeholders, including government, private sector entities, and civil society organizations. Additionally, these green financing mechanisms must be in line with the more general aims and objectives of sustainable development.

In this regard, the Fuzzy Multi Criteria Decision-Making (MCDM) methods are useful tool for evaluating green financing mechanisms for natural resource management. This approach allows decision-makers to take into consideration the imprecise or ambiguous information that is commonly associated with natural resource management, which can lead to better-informed judgments and more effective allocation of resources (Kubler et al., 2016). The Fuzzy Analytical Hierarchy Process (AHP) method is employed to assess the criteria and sub-criteria that are pertinent to sustainable natural resource management and the SDGs in China. While, the Fuzzy VlseKriterijumska Optimizcija I Kaompromisno Resenje (VIKOR) approach is used to prioritize the critical green financing mechanisms to achieve sustainable natural resource management targets. Thus, China's natural resource management and environmental protection greatly benefit from the green financing mechanisms. These finance mechanisms can offer the funds required to support initiatives that advance environmental protection, foster economic growth, and support sustainable development. To have the greatest impact, these financing mechanisms should be adapted to country's particular needs and characteristics as well as integrated into more comprehensive frameworks and policies for sustainable development.

Furthermore, our research objectives are to evaluate the role of green finance in managing natural resources in China to provide essential insights that will guide policy formulation and execution. It seeks to address the knowledge and policy gaps in green finance in the Chinese context while providing valuable inputs to stakeholders looking to adopt green finance practices for sustainable resource management in China. By delving into the practical implications of green financing strategies, it aims to enhance the understanding of green finance and its applicability in natural resource management, significantly contributing to the existing body of knowledge. By focusing on China, this study offers critical insights that could inform policymaking in similar contexts worldwide while highlighting the need for multidimensional, innovative approaches such as green finance for addressing the sustainability challenges of the 21st century.

2. Theoretical background

The literature review focuses on the importance of natural resource management and the role of green financing mechanisms in achieving SDGs in China. The Chinese government has put in place a number of laws and programs to address these issues because it understands the value of managing natural resources. The "Ecological Civilization Policy", one of the measures adopted by the Chinese government, attempts to advance sustainable development by striking a balance between economic development and environmental preservation (Mi et al., 2022). By increasing resource efficiency, lowering pollution, and safeguarding biodiversity, the plan attempts to promote green development and highlights the significance of natural resource management. The government has also implemented various initiatives to support the management of natural resources, such as the "Green for Grain" initiative, which encourages farmers to turn farmland into forested and grassland in exchange for cash rewards (Fan and Xiao, 2020). The program has been effective at encouraging sustainable land use methods and minimizing soil erosion. Environmental and natural resource management were also affected by the COVID-19 pandemic. The pandemic impacted supply systems, consumer patterns, and resource use. Lockdowns and limitations during the pandemic temporarily lowered pollution and greenhouse gas emissions due to reduced industrial and transportation activity. These unforeseen environmental advantages showed that human behaviors affect the environment (Ali Shah et al., 2021). The worldwide outbreak also revealed gaps in global supply networks for vital commodities, stressing the need for resource resilience and sustainable sourcing to achieve SDGs.

China's efforts toward sustainable natural resource management have triggered a growing body of research, policymakers, and stakeholder engagement, emphasizing the potential role of green financing mechanisms (Feng and Yang, 2023; Geng et al., 2023). Scholars and policymakers strive to understand how these financial strategies can be harnessed for environmental sustainability and meeting the SDGs, making this issue a central theme in contemporary academic and policy debates (Jiakui et al., 2023; Lee and Lee, 2022). The existing literature review shows a growing recognition of the interconnection between green financing mechanisms and sustainable resource management.

Various theoretical and empirical studies deepen the understanding of the role and effectiveness of green financing mechanisms in promoting sustainable development and natural resource management in China. For instance, studies such as (Khan et al., 2022), which examined the determinants of China's environmental quality, and (Khan and Hou, 2021a) investigating the relationship between energy consumption, tourism growth, and the ecological footprint in IEA countries, provided crucial insight into the broader context within which these mechanisms operate. This understanding was essential for the subsequent evaluation of green financing strategies within the Chinese context. (Chen et al., 2023) similarly underscored the importance of energy efficiency and economic prosperity within a sustainable environment, providing further depth to our understanding of the challenges and opportunities associated with adopting and implementing green financing mechanisms. (Zakari et al., 2023) also contributed to this understanding, focusing on the impact of environmental technology innovation and energy credit rebates on carbon emissions. Another study by (Khan and Hou, 2021b) discussing the role of multilateral environmental diplomacy in improving environmental quality, particularly with the United

Table 1
The proposed criteria and sub-criteria of this study.

Criteria	Sub-criteria	Short summary
Stakeholder participation	Community empowerment	The effectiveness of natural resource management programs to involve local populations in decision-making depends on this sub-criterion.
	Marginalized group participation	To ensure participation by underrepresented groups including women, indigenous people, and small-scale farmers.
	Transparency and accountability	This sub-criteria is crucial for considering if programs for natural resource management can promote accountability and openness.
Financial sustainability	Private investment	This sub-criteria is important for determining if projects for natural resource management have a chance of attracting private investment.
	Cost-effectiveness	This sub-criteria is important for ensuring that natural resource management initiatives are economical and can create enough income to maintain themselves over the long run.
	Innovative financing mechanisms	This sub-criterion is relevant to evaluating the potential for revenue generation through green finance mechanisms such as green bonds and carbon markets.
Environmental impact	Greenhouse gases (GHGs) emission	This sub-criterion is relevant to determining the possibility for natural resource management programs to minimize GHGs emission.
	Sustainable land use	The feasibility of implementing sustainable land use techniques is an important consideration when evaluating initiatives in the field of natural resource management.
	Biodiversity conservation	This sub-criteria can be used to assess how well natural resource management initiatives preserve biodiversity and protect endangered species.
Social equity	Livelihood improvement	This sub-criteria is crucial for considering if natural resource management programs have the ability to raise the standard of living in the surrounding areas.
	Employment opportunities	This criterion is essential for figuring out if natural resource management projects would boost local employment.
	Gender equality	This criterion is essential for assessing the potential of natural resource management strategies to promote gender equality and empower women.
Technological feasibility	Local resource compatibility	This sub-criteria is important for evaluating how natural resource management initiatives fit with the conditions and resources available locally.
	Scalability	This sub-criteria determines if natural resource management strategies can scale and expand.
	Innovative technologies	The ability of natural resource management initiatives to use cutting-edge and ecologically friendly technologies can be assessed using this sub-criteria.
Institutional feasibility	Public and private sector support	The success of projects involving the management of natural resources depends on their ability to gain support from the public and private sectors.
	Policy alignment	This sub-criteria can be used to evaluate how well national policies and regulations align with projects for managing natural resources.
	Stakeholder partnerships	This sub-criteria is important for evaluating the potential for partnerships to form amongst various stakeholders, such as government entities, civil society groups, and corporate sector actors.

States, gave an international perspective and a comparative understanding of the issues and solutions related to environmental sustainability and green financing. Finally, (Liu et al., 2023) examined the path to sustainable development and emphasized the importance of environmental taxes and governance, which certainly have implications for green financing mechanisms. However, despite the focus on these initiatives, empirical studies have seldom validated their effectiveness and impact within China's unique socio-economic and ecological context.

Therefore, using the Fuzzy MCDM approach has helped evaluate green financing options, given its proficiency in handling ambiguity and uncertainty inherent in sustainable natural resource management decisions (Ahmed et al., 2020). In contrast, previous studies have applied this approach in diverse settings, such as assessing the viability of green bonds for financing renewable energy projects (Rasoulinezhad and Taghizadeh-Hesary, 2022) and determining the applicability of green financing for sustainable agriculture (Mo et al., 2023). However, there are still research gaps to be addressed. (Jaber et al., 2021) warn that neglecting contextual and institutional factors while using the Fuzzy MCDM approach should be avoided. This suggests that future research should incorporate these elements to provide a more accurate and nuanced understanding of green financing mechanisms.

Supporting this argument, (Hailiang et al., 2023) underline the need for more empirical studies to validate the theoretical link between green financing and sustainable development, especially regarding China's progress toward the SDGs. This plea for more robust and empirically grounded research mirrors a broader call for evidence-based policy-making within the green finance sector. Another critical aspect overlooked mainly in the existing literature is the role of institutional factors in shaping the implementation and effectiveness of green financing mechanisms. A study by (Yin and Xu, 2022) illustrates how existing institutional structures can either facilitate or impede the adoption and impact of green financing strategies.

Stakeholder involvement need to evaluate how the COVID-19 pandemic and conflicts affect green finance provisions in China. Due to the pandemic's rapid economic slump and uncertainty, governments and financial institutions globally reassessed their plans and goals. China's COVID-19 reaction supported economic recovery and sustainable growth, which affected green financing. Fresh circumstances forced financial sector stakeholders to adapt, making it important to investigate how these developments affected green project and initiative financing. Domestic and international conflicts can affect financial markets and investor confidence. Conflicts may increase risks, discouraging private investors from green finance. Geopolitical issues, security concerns, and objectives associated with green finance need to be addressed. In a fast-changing world, understanding how conflicts may enable or hamper stakeholders to prioritize green financing mechanisms is essential for effective strategies. Stakeholders shape green financing structures, and external influences like the COVID-19 epidemic and conflicts can affect stakeholder attitudes and encouragement. These events affect green financing in China, consequently adaptable and resilient ways to meet SDGs must be developed. This study uses Fuzzy MCDM to unravel these complex dynamics and improve our understanding of green financing in the face evolving challenges.

2.1. Identification of the criteria and sub-criteria relevant to SDGs and natural resource management in China

Evaluation and prioritization of green financing mechanisms require a thorough understanding of the criteria and sub-criteria that apply to China's SDGs and natural resource management. The multiple relevant criteria and sub-criteria for managing China's natural resources and achieving the SDGs is given in Table 1.

The effectiveness, sustainability, and SDG compliance of China's natural resource management activities require consideration for external factors like the COVID-19 plague. Pandemic has shown the necessity for adaptable and resilient sustainability targets. Thus, when assessing green financing methods against these criteria and sub-criteria, resilience to unexpected interruptions like COVID-19 must be considered. This approach will assist identify strategies that comply with immediate goals while demonstrating the flexibility and resilience to cope with complex and uncertain global situations.

2.2. Identification of the green financing mechanisms relevant to SDGs and natural resource management in China

It is important to recognize and describe the green financing strategies to manage China's natural resources and accomplish the SDGs. In this study, the several key financing strategies have been identified from the literature review. In this regard, the country can promote the SDGs and resource management through the following identified green financial strategies.

2.2.1. Green bonds

To finance environmentally friendly projects, "green bonds" are a type of debt instrument (Bhutta et al., 2022). Green bonds can assist China in managing its natural resources by providing a new source of financing for activities involving sustainable agriculture, renewable energy, and energy efficiency. The transition to a green economy can also be supported via green bonds, which can encourage private investment (Xiang and Cao, 2023).

2.2.2. Green investment funds

Funds that invest in businesses or projects that improve the environment are known as "green investment funds." By providing a fresh source of funding for initiatives related to sustainable agriculture, renewable energy, and energy efficiency, green investment funds can improve China's management of its natural resources. By attracting private resources, green investment funds can also aid in the transition to a green economy (Dawid et al., 2021).

2.2.3. Public-private partnerships

Public-private partnerships bring these sectors together to deliver public services or infrastructure projects. Public-private partnerships can help in the management of natural resources in China by pooling their resources and expertise to achieve sustainable development objectives. Public-private partnerships can also encourage private investment growth and innovation in the management of natural resources (Lu et al., 2022).

2.2.4. Carbon markets

Companies can buy and sell carbon credits on carbon markets to reach their emission reduction goals. Projects for managing natural resources, such afforestation and reforestation, can be financed with the fund raised by carbon markets (Kong and Wang, 2022; Moser et al., 2022). By encouraging businesses to cut their GHGs emission and support the transition to a low-carbon economy, carbon markets can also support sustainable development.

2.2.5. Microfinance

Small loans are given to people and small enterprises as part of microfinance. By providing funding for small-scale initiatives related to sustainable agriculture, renewable energy, and eco-friendly tourism, microfinance can aid in the management of natural resources in China. Microfinance can also support in the reduction of poverty and local entrepreneurship to forward the cause of sustainable development (Gatto and Sadik-Zada, 2022).

2.2.6. Sustainable tourism financing

Financing for sustainable tourism includes funding initiatives that advance the industry and safeguard the environment (Xu et al., 2020). By providing finance for ecotourism, sustainable transportation, and waste management projects can help in China's natural resource management. By offering green financing sources, encouraging private sector investment, and assisting the shift to a green economy, these green financing structures can aid in natural resource management and the achievement of SDGs in China.

3. Methods

The principles of fuzzy logic are used in hybrid MCDM systems to tackle decision-making issues involving imprecise or uncertain information (Kahraman et al., 2015). In this regard, this study utilizes Fuzzy AHP and Fuzzy VIKOR approaches to assess the criteria, sub-criteria, and green financing mechanism for natural resource management and implications for achieving SDGs in China. A general methodology structure based on hybrid fuzzy MCDM methods is presented in Fig. 1.

3.1. The fuzzy AHP method

In the 1970s, Saaty proposed the AHP approach (Saaty, 1990). Since it can combine qualitative and quantitative data into a single model, the AHP is a crucial MCDM methodology (Solangi et al., 2019b; Wang et al., 2020). However, the Fuzzy based AHP approach is applied in this study to produce more meaningful results. Because fuzzy set theory can be applied when the decision problem is only vaguely, imperfectly, or uncertainly known (Dai and Solangi, 2023; Krejčí, 2018). Triangular Fuzzy Numbers (TFNs) are then used to operate the pairwise comparison into a matrix. Table 2 displays the TFNs rating scale that was used in this investigation.

The key steps of the Fuzzy AHP method are defined as follows (Almalki Sultan Musaad et al., 2020).

Step 1 Construct the pairwise comparison matrix:

Step 2 Describe the fuzzy synthetic extent value AB_i:

$$AB_i = \sum_{j=1}^{m} T_{gi}^{j} \times \left[\sum\nolimits_{i=1}^{n} \sum_{j=1}^{m} T_{gi}^{j} \right]^{-1}$$

$$\sum_{j=1}^{m} T_{gi}^{j} = \left(\sum_{j=1}^{m} x1_{ij}, \sum_{j=1}^{m} x2_{ij}, \sum_{j=1}^{m} x3_{ij} \right)$$

$$\left[\sum_{i=1}^{n} \sum_{j=1}^{m} T_{gi}^{j} \right]^{-1} = \left(\frac{1}{\sum_{n}^{i=1} \sum_{m}^{j=1} x3_{ij}}, \frac{1}{\sum_{n}^{i=1} \sum_{m}^{j=1} x2_{ij}}, \frac{1}{\sum_{n}^{i=1} \sum_{m}^{j=1} x1_{ij}} \right) \tag{1}$$

Step 3 Compare the obtained values of AB_i:

$$V(AB_j \geq AB_i) = (X) = \begin{Bmatrix} 1, \text{in} \\ \text{case of} x2_j \geq x2_i \\ 0, \text{in} \\ \text{case of} x1_i \geq x3_j \\ \dfrac{x1_i - x3_j}{(x2_j - x3_j) - (x2_i - x1_i)}, \text{otherwise} \end{Bmatrix} \tag{2}$$

where X indicates the highest point between AB_j and AB_i.

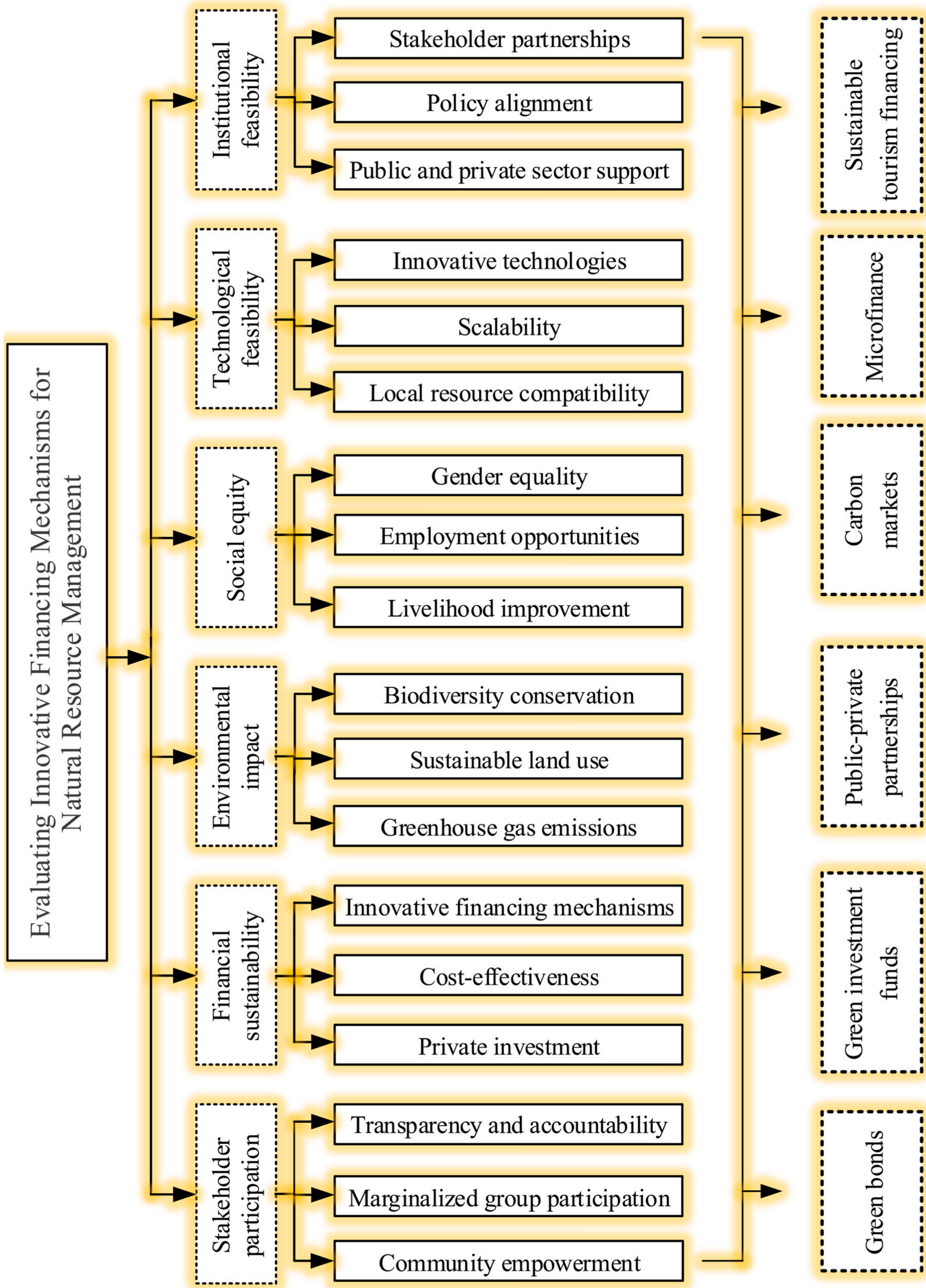

Fig. 1. The hierarchical structured methodology.

Table 2
Linguistic preference based TFNs scale.

Number	Linguistic Variables	TFNs
1	Equally preferred	(1,1,1)
2	Equally to moderately preferred	(1,2,3)
3	Moderately preferred	(2,3,4)
4	Moderately to strongly preferred	(3,4,5)
5	Strongly preferred	(4,5,6)
6	Strongly to very strongly preferred	(5,6,7)
7	Very strongly preferred	(6,7,8)
8	Very strongly to extremely preferred	(7,8,9)
9	Extremely preferred	(8,9,10)

Step 4 Analyze the minimum possibility degree $X(i)$ of $V\ (AB_j \geq AB_i)$ for $(ij = 1,2,3,4,5,\dots,k)$:

$$\begin{gathered} V(AB \geq AB_1, AB_2, AB_3, AB_4, AB_5\dots, AB_k), \\ \text{for } (i = 1,2,3,4,5\dots,k) = V[(AB \geq AB_1),(AB \geq AB_2), \text{and}\dots(AB \geq AB_k)] = \min V(AB \geq AB_i) \\ \text{for } (i = 1,2,3,4,5,\dots,k) \end{gathered} \tag{3}$$

Step 5 Compute the weight vector:

$$W' = (X'(Y_1), X'(Y_2), X'(Y_3), X'(Y_4), X(Y_5), \dots, X'(Y_n))^T \tag{4}$$

Finally, the weight vector can be normalized as:

$$W = (d(A_1), d(A_2), d(A_3), d(A_4), d(A_5), \dots, d(A_n))^T \tag{5}$$

where W shows a non-fuzzy number.

3.2. The fuzzy VIKOR method

In 1980, Opricovic created the VIKOR method (Opricovic and Tzeng, 2004). This approach prioritizes the available options and finds reasonable solutions to a decision-problem with contradicting or competing criteria, which might aid decision-makers in making a final choice (Solangi et al., 2019a). A compromise is an agreement reached through mutual concessions, and in this case it is a feasible solution that comes the closest to the ideal. In this study, the fuzzy-based VIKOR approach is used to solve the decision-making problem's uncertainty. Table 3 displays the TFNs rating scale that is widely used in MCDM issues.

The key steps of Fuzzy VIKOR methods are as follows (Nezami and Yildirim, 2013).

Step 1 Establish the fuzzy performance matrix and the weight vector

Table 3
TFNs scale.

Number	Linguistic Variables	TFNs
1	Very poor	(0,0.05,0.15)
2	Poor	(0.1,0.2,0.3)
3	Fairly Poor	(0.2,0.35,0.5)
4	Fair	(0.3,0.5,0.7)
5	Fairly Good	(0.5,0.65,0.8)
6	Very Good	(0.7,0.8,0.9)
7	Extremely Good	(0.85,0.95,1)

$$\widetilde{D} = \begin{array}{c} \\ O_1 \\ \vdots \\ O_n \end{array} \begin{bmatrix} C_1 \ \ C_2 \cdots C_n \\ \widetilde{p}_{11}\widetilde{p}_{12}\cdots\widetilde{p}_{1n} \\ \widetilde{p}_{21}\widetilde{p}_{22}\cdots\widetilde{p}_{2n} \\ \vdots\vdots\vdots\vdots \\ \widetilde{p}_{m1}\widetilde{p}_{m2}\cdots\widetilde{p}_{mn} \end{bmatrix} \tag{6}$$

$$\widetilde{W} = (w_1, w_2, w_3), \sum_{j=1}^{n} w_j = 1$$

where O_i represents the alternative i, i.e., $\text{i} = (1,2,3,\dots,\text{m})$; C_j denotes the criteria j, i.e., $\text{j} = (1,2,3,\dots,\text{n})$.

Step 2 Establish the values of all benefit criteria and the cost criteria.

$$\widetilde{p}_i^+ = max_j\ \widetilde{p}_{ij}, \widetilde{p}_i^- = min_j \widetilde{p}_{ij} \text{ for } i \in I^b$$

$$\widetilde{p}_i^+ = min_j\ \widetilde{p}_{ij}, \widetilde{p}_i^- = max_j \widetilde{p}_{ij} \text{ for } i \in I^c \tag{7}$$

Step 3 Compute the normalized fuzzy decision matrix $\widetilde{D}_{ij}$:

$$\widetilde{D}_{ij} = \frac{\widetilde{p}_i^+(-)\widetilde{p}_{ij}}{z_i^+ - l_i^-} \text{ for } i \in I^b$$

$$\widetilde{D}_{ij} = \frac{\widetilde{p}_{ij}(-)\widetilde{p}_i^+}{z_i^- - l_i^+} \text{ for } i \in I^c \tag{8}$$

Step 4 Compute the values $\widetilde{S}_j = (\widetilde{S}_j^x, \widetilde{S}_j^y, \widetilde{S}_j^z)$ and $\widetilde{R}_j = (\widetilde{R}_j^x, \widetilde{R}_j^y, \widetilde{R}_j^z)$:

$$\widetilde{S}_j = \sum_{i=1}^{n} \widetilde{W}_i(\times)\widetilde{D}_{ij} \tag{9}$$

$$\widetilde{R}_j = max_i \widetilde{W}_i(\times)\widetilde{D}_{ij} \tag{10}$$

Step 5 Compute the values $\widetilde{Q}_j = (\widetilde{Q}_j^x, \widetilde{Q}_j^y, \widetilde{Q}_j^z)$:

$$\widetilde{Q}_j = v\frac{\widetilde{S}_j(-)\widetilde{S}^+}{S^{-z} - S^{+x}}(+)(1-v)\frac{\widetilde{R}_j(-)\widetilde{R}^+}{R^{-z} - R^{+x}} \tag{11}$$

here $\widetilde{S}^+ = min_j\ \widetilde{S}_j;\ S^{-z} = max_j S_j^z;\ \widetilde{R}^+ = min_j \widetilde{R}_j;\ R^{-z} = max_j R_j^x$

Moreover, v is presented as a group strategy weight for the majority of criteria $\widetilde{S}_j$; whereas, $(1-v)$ is the individual weight of $\widetilde{R}_j$.

Step 6 & 7: Defuzzify $\widetilde{S}_j$, $\widetilde{R}_j$ and $\widetilde{Q}_j$ values

Step 8 Propose a solution to the alternative $O^{(1)}$ which is the optimal solution of the measure Q, if the two conditions are satisfied.

The complex environment of sustainable development in China requires knowledge from a broad range of stakeholders, especially when undertaking a Fuzzy MCDM study to analyze and prioritize green finance sources for natural resource management. Since outbreak brought up unanticipated issues and potential for sustainable development, the COVID-19 pandemic has highlighted the importance for multi-disciplinary collaboration. The study emailed six experts a questionnaire. These expertise are excellent for a Fuzzy MCDM study reviewing and prioritizing green funding mechanisms for natural resource management in China to accomplish SDGs. China's natural resource management specialists understand sustainable development's potential and limitations. Chinese policy and law professionals can better understand China's legal and regulatory framework for natural resource management and sustainable development. This study includes academics, practitioners, and policymakers because of their complementing knowledge and skills. This would aid a fair assessment of China's resource management green funding options.

4. Results and analysis

4.1. Ranking of criteria using fuzzy AHP method

The criteria findings were acquired using the Fuzzy AHP approach. Table 4 lists the rankings and weights of the various criteria. The ranking shows that financial sustainability having the greatest importance (0.192), making it the most significant factor to consider when evaluating and ranking green financing mechanisms. As a result, it is suggested that the financing procedures should be commercially successful, financially sound, and capable of luring resources and investments that support China's efforts to develop sustainably and manage its natural resources. With a second-highest weight of 0.176, institutional viability is a crucial factor to consider when evaluating and ranking alternative financing strategies. This means that the financing mechanisms ought to be in line with China's institutional and legal frameworks and ought to have the backing of government institutions, civil society groups, and other interested parties. Environmental effect is a crucial factor to consider when evaluating and ranking green finance mechanisms because it has the third-highest weight (0.171). This implies that the green finance ought to support environmental sustainability and to aid in the preservation and restoration of China's natural resources. Lower weights for stakeholder involvement, technological viability, and social equality show that these factors are less crucial when evaluating and ranking green finance mechanisms.

4.2. Ranking of sub-criteria with respect to stakeholder participation

Fig. 2 presents the ranking of sub-criteria associated with stakeholder participation. The sub-criteria for transparency and accountability have the greatest weight, with a weight of 0.356, indicating that they are the most significant sub-criteria in evaluating and prioritizing green financing strategies for natural resource management in China in order to achieve SDGs. This implies that the financing mechanisms should be accountable to stakeholders, such as government organizations, civil society groups, and local communities, and transparent in their operations, decision-making processes, and resource allocation. This ensures that financial structures support natural resource management plans that benefit all stakeholders and are sustainable.

Community empowerment is the sub-criteria with the second-highest weight, at 0.331, showing its significance in evaluating and ranking green finance systems. This argues that local communities in China should have more access to green financing mechanisms so they may participate in decision-making and gain from sustainable development and resource management techniques. The marginalized group involvement sub-criteria have the lowest weight (0.312), which suggests that it is less significant when analyzing the green finance strategies. The inclusion of marginalized groups, such as women, children, and indigenous peoples, in the decision-making process as well as in the advantages of sustainable development and natural resource management practices in China should be ensured. Overall, the analysis points to green financing mechanisms that are better fit for China's natural resource management and SDG achievement.

Table 4
The ranking of main-criteria.

Criteria	Weight	Rank
Stakeholder participation	0.165	4
Financial sustainability	0.192	1
Environmental impact	0.171	3
Social equity	0.143	6
Technological feasibility	0.152	5
Institutional feasibility	0.176	2

4.3. Ranking of sub-criteria with respect to financial sustainability

Fig. 3 displays the ranking of sub-criteria related with financial sustainability. The sub-criteria for green financing mechanisms for natural resource management in China have the highest weight, with a weight of 0.348, indicating that they are the most significant sub-criteria in assessing and prioritizing such financing mechanisms in order to achieve SDGs. This means that the financing methods should be creative and capable of attracting resources and investments that support China's natural resource management policies and practices for sustainable development. Green bonds, public-private partnerships, and carbon markets are a few examples of green financing strategies that might open up new avenues for luring capital and resources to support sustainable growth and good resource management practices in China. Cost-effectiveness is the sub-criteria with the second-highest weight, at 0.334, showing its significance in evaluating and ranking green financing systems. This shows that in order to achieve sustainable development and natural resource management practices in China, the financing mechanisms should be cost-effective, efficient, and deliver value for money. Private investment is the sub-criteria with the lowest weight (0.317), indicating that it is less significant in evaluating the green financing strategies. This sub-criterion should not be disregarded, either, as private investment can assist sustainable growth and methods for resource management in China.

4.4. Ranking of sub-criteria with respect to environmental impact

Fig. 4 shows the ranking of sub-criteria linked with environmental impact. The sub-criterion of GHGs emission, which has the highest weight of 0.352, is the most significant sub-criterion when evaluating and giving priority to green finance mechanisms for natural resource management in China in order to achieve SDGs. This implies that decreasing GHGs emission and accelerating the shift to a low-carbon economy should be the main goals of the finance systems. Investments in clean technologies, efficient energy use, and renewable energy sources can all help China achieve sustainable growth and effective resource management while lowering GHGs emission. With a weight of 0.332, the sub-criteria for biodiversity protection ranks second in importance, demonstrating its significance in evaluating and ranking green finance mechanisms. This implies that China's funding structures should support the preservation and restoration of the country's biodiversity and ecosystems. This can involve spending on projects that help China's biodiversity be protected and improved, such as habitat restoration, species conservation, and sustainable forestry methods. Sustainable land use is the sub-criteria with the lowest weight (0.315), indicating that it is less significant for evaluating and ranking green finance strategies. Sustainable land use strategies can support China's efforts at sustainable development and resource management, thus it is important not to disregard this sub-criterion.

4.5. Ranking of sub-criteria with respect to social equity

Fig. 5 depicts the ranking of sub-criteria with respect to social equity. The sub-criterion of gender equality has the highest weight, with a weight of 0.341, indicating that it is the most significant sub-criterion when prioritizing green financing mechanisms for natural resource management to accomplish SDGs. This recommends that the green

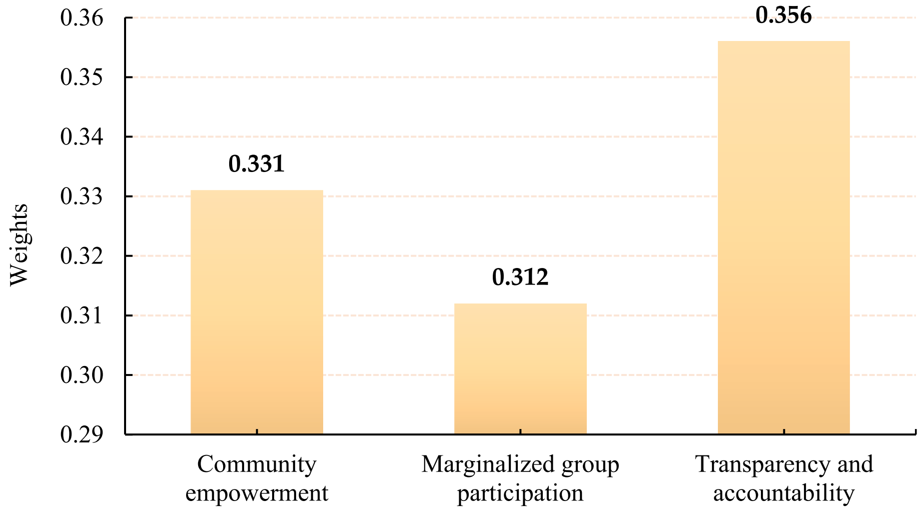

Fig. 2. The ranking of sub-criteria (stakeholder participation).

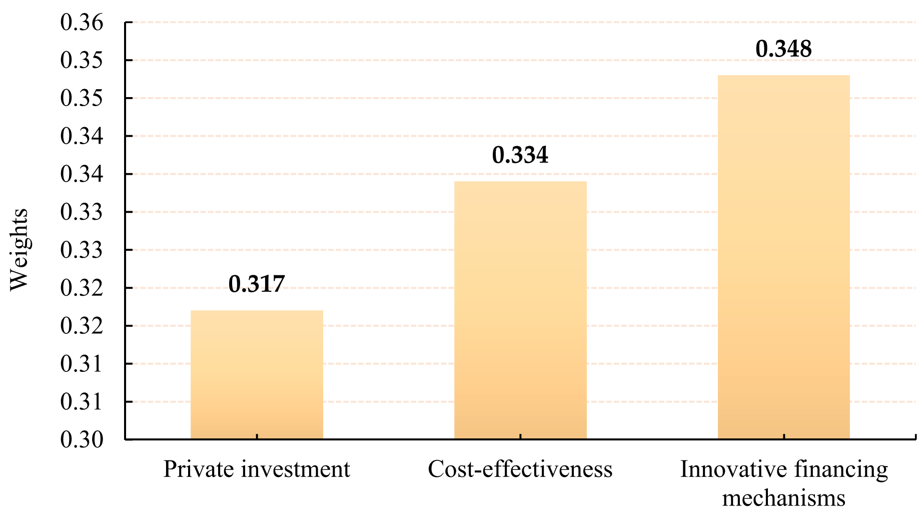

Fig. 3. The ranking of sub-criteria (financial sustainability).

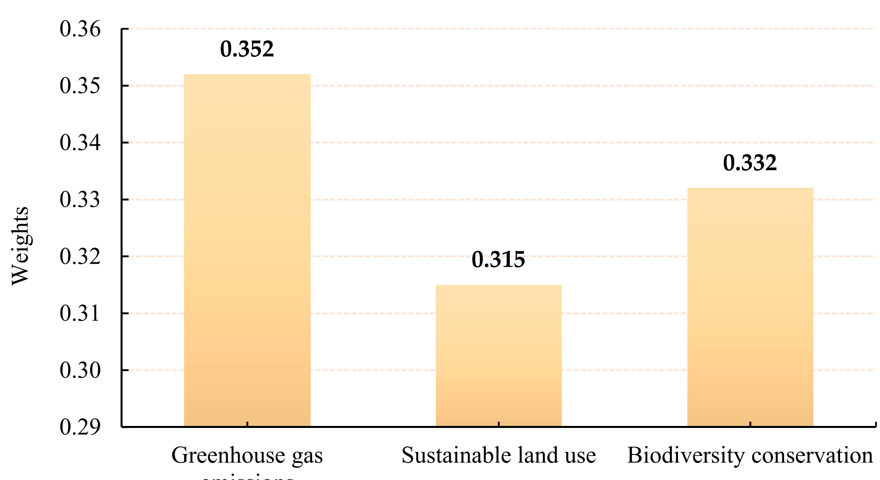

Fig. 4. The ranking of sub-criteria (environmental impact).

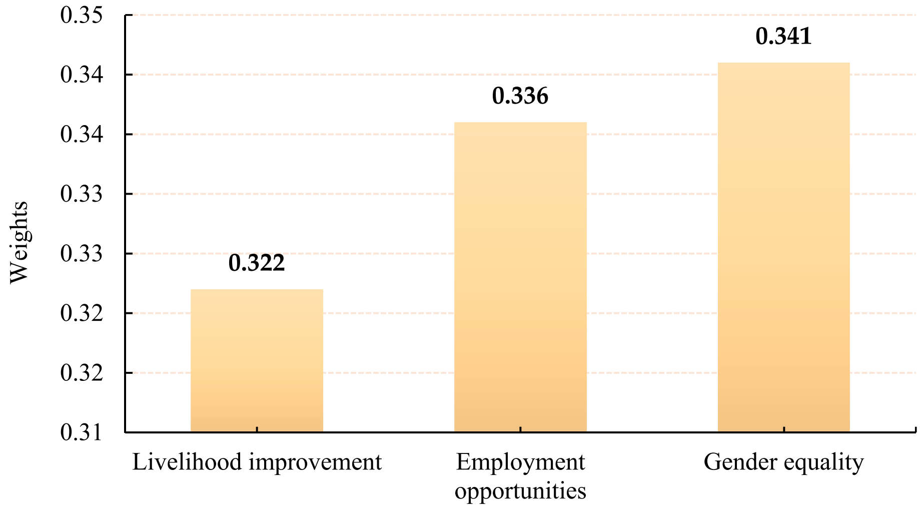

Fig. 5. The ranking of sub-criteria (social equity).

finance approaches should support women's empowerment and gender equality, giving both men and women equal opportunity to participate in decision-making and gain from China's sustainable development and resource management techniques. This might involve financing for initiatives that empower women through education and training, as well as encouraging women to take on roles of leadership and decision-making.

With a weight of 0.336, the sub-criteria for job opportunities ranks second in importance, demonstrating its significance in evaluating and ranking green finance mechanisms. This implies that employment possibilities for local communities, especially those who may have been historically disenfranchised, should be produced by the finance channels. Investing in green jobs, sustainable agriculture, and tourism can help to enhance local communities' standard of living in China while also opening up job opportunities. With the lowest weight of 0.322, the sub-criteria for livelihood improvement is less significant when evaluating and ranking green financing mechanisms.

4.6. Ranking of sub-criteria with respect to technological feasibility

Fig. 6 presents the ranking of sub-criteria with respect to technological feasibility. According to the greatest weight, the local resource compatibility sub-criteria has the highest weight of 0.363, indicating that it is the most significant sub-criteria in evaluating and selecting green finance methods for natural resource management in China in order to realize SDGs. This means that the green financing strategies used in China should be suitable with the country's unique conditions and resources, including the availability of natural resources, local customs and cultural norms, and the country's political and economic environment. This can guarantee that the finance mechanisms are appropriate, applicable, and sustainable over the long term in the local context.

Scalability is the sub-criteria with the second-highest weight (0.329), indicating that it is crucial for assessing the green finance systems. This implies that the finance mechanisms need to be scalable and flexible enough to be expanded to cover more territory or a larger population. Due to this, China's sustainable development and natural resource management methods can benefit more from the finance arrangements. With the lowest weight of 0.307, the sub-criteria for novel technology is less significant when evaluating and ranking green financing mechanisms. Innovative technologies can significantly contribute to the promotion of sustainable development and natural resource management methods in China, hence this sub-criterion should not be disregarded.

4.7. Ranking of sub-criteria with respect to institutional feasibility

Fig. 7 demonstrates the ranking of sub-criteria with respect to institutional feasibility. The sub-criteria of policy alignment have the highest weight, 0.375, according to the highest weight, indicating that it

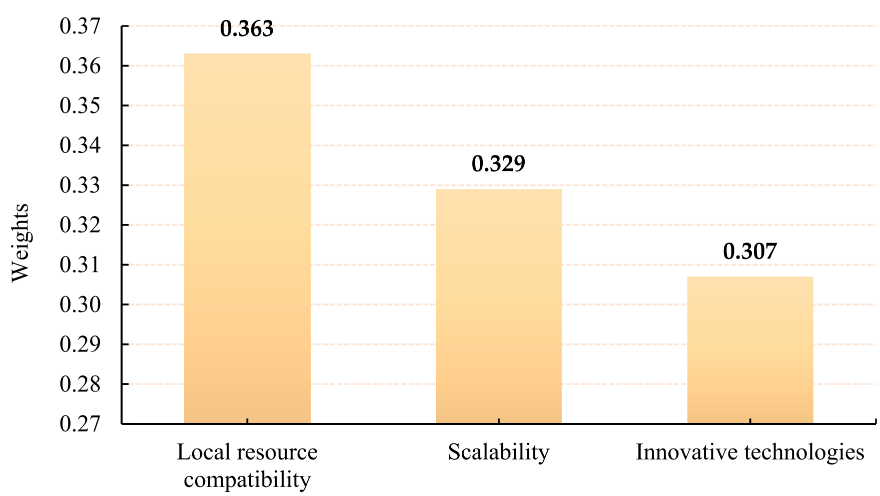

Fig. 6. The ranking of sub-criteria (technological feasibility).

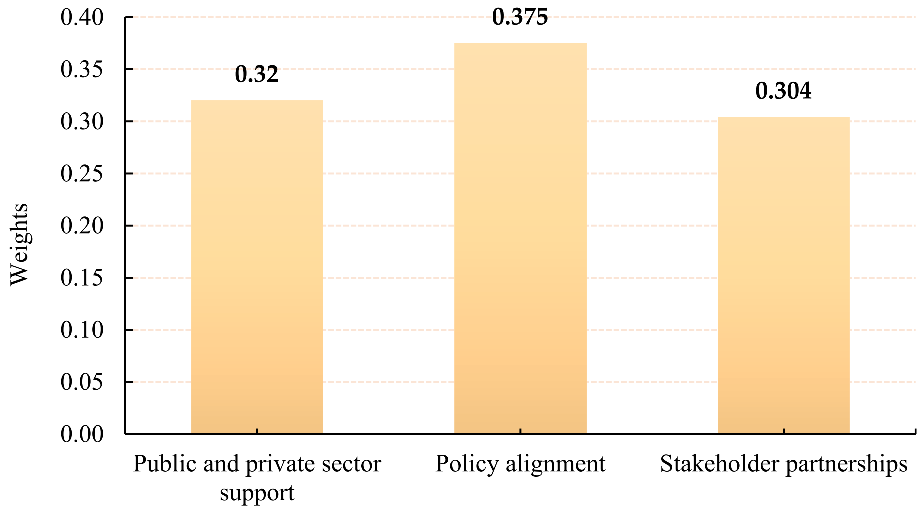

Fig. 7. The ranking of sub-criteria (institutional feasibility).

is the most substantial sub-criteria in evaluating and giving priority to green finance methods for natural resource management to achieve SDGs. This indicates that financing arrangements should comply with relevant Chinese policies and legislation, such as sustainable development and resource management. This allows China's policy framework for sustainable development and natural resource management to support financial mechanisms legally and institutionally.

Public and private sector support is a sub-criterion with the second-highest weight (0.320), indicating that it is crucial for assessing the green financing strategies. This recommends that both the public and private sectors in China, including government organizations, private investors, and civil society organizations, should support the financing structures. This can guarantee that the finance mechanisms have the assets and know-how required to effectively advance sustainable development and methods of resource management in China. Stakeholder partnerships as a sub-criteria have the lowest weight (0.304), indicating that they are less significant when evaluating and ranking green financing strategies. Stakeholder collaborations, however, can be crucial in advancing sustainable development and natural resource management techniques in China, therefore this sub-criterion should not be disregarded.

4.8. *Results of financing mechanisms (alternatives) using fuzzy VIKOR method*

The results of green financing approaches for natural resource management were obtained utilizing the Fuzzy VIKOR approach in this part. The S, R, and Q values of green finance strategies are shown in Table 5. The lowest value of Q is regarded as the best choice for China's natural resource management and the achievement of the SDGs. The ultimate ranking based on Q value is displayed in Fig. 8.

According to the findings, carbon markets are the best financing option for China's SDGs and effective resource management. This implies that carbon markets can offer a market-based mechanism to encourage emission reduction and resource management practices in

Table 5
The ranking of financing mechanisms based on S, R, and Q values.

Green financing mechanisms	S_i	R_i	Q_i	Rank
Green bonds	0.165	0.272	0.038	0.066
Green investment funds	0.192	0.287	0.038	0.074
Public-private partnerships	0.171	0.258	0.043	0.100
Carbon markets	0.143	0.209	0.041	0.062
Microfinance	0.152	0.284	0.042	0.104
Sustainable tourism financing	0.176	0.274	0.044	0.113

China, while also producing income that can be utilized to support sustainable development efforts. The second-highest and most suitable financing method for China's SDGs and natural resource management is provided by green bonds. Green bonds can fund sustainable development projects like natural resource management while making investors money. Green investment funds finance resource management and sustainable growth. Green investment funds can finance sustainable development projects like China's natural resource management plans with institutional and individual investors.

In China, public-private partnerships are recognized as the fourth best financing option for meeting the SDGs and managing natural resources. Public-Private Partnerships can be efficient at utilizing the resources and knowledge of the private sector, but they might run into issues with transparency and accountability. For China's SDGs and natural resource management, microfinance is a less suitable financing channel. Microfinance can give small businesses and entrepreneurs access to capital, but it might not be enough to fund significant sustainable development projects. The least weighted funding instrument for SDGs and natural resource management in China is sustainable tourism, which also has the lowest weight. Even though sustainable tourism can benefit local economies and resource management, it might not bring in enough money to support sustainable development projects.

The analysis generally suggests that carbon markets and green bonds are the best financing options for China's implementation of SDGs and natural resource management. These green finance options can support sustainable development and natural resource management practices in the country.

4.9. *Discussion*

In assessing and prioritizing green financing options for natural resource management in China, external factors like the COVID-19 epidemic may affect their importance and effectiveness. The pandemic has changed economic landscapes and sustainability priorities, which may affect green financing evaluation criteria and sub-criteria. In this regard, the findings show that a number of criteria and sub-criteria are significant when evaluating and ranking green financing mechanisms for natural resource management in China. Both the Fuzzy AHP and Fuzzy VIKOR methodologies were used to examine the multiple criterion, sub-criteria, and the green financing mechanisms in the context of China. The financial sustainability, institutional feasibility, and environmental impact are the key criteria of this study. This indicates that achieving SDGs and effective natural resource management in China will be more feasible with green finance structures that put an emphasis on these criteria. Further findings reveal that the carbon markets and

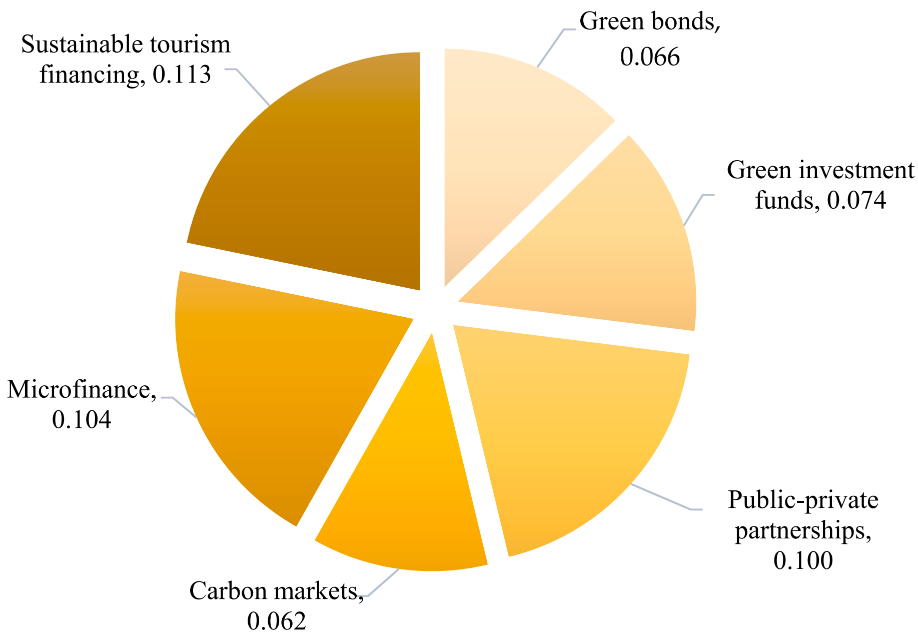

Fig. 8. The final ranking of green financing strategies based on lowest value of Q.

green bonds are most significant green financing strategies for accomplishing SDGs and natural resource management practices in China.

This study supports previous studies that emphasize social justice and environmental sustainability for sustainable development and resource management. In earlier study, the authors found that environmental sustainability was essential to sustainable development in China (Chen et al., 2022). The recent study found that China's environmental sustainability requires addressing GHGs emission, biodiversity, and land usage. Previous studies have stressed the need of social inclusion and gender equality in China's sustainable development policies (Lee, 2020). Promoting gender equality and women's participation in decision-making and access to school and job was essential to sustainable development in China. This study highlights the importance of institutional and technological viability in promoting sustainable development and natural resource management in China. Moreover, the previous researches (Ibrahim et al., 2022; Larsen, 2022; Sachs et al., 2019) identified that financial, institutional, environmental, and technological feasibility were key to sustainable urban development in China, with an emphasis on supporting efficient green finance, governance structures, and adopting cutting-edge technology to increase resource efficiency.

A study conducted by (Xiang and Cao, 2023) discovered that green bonds can serve as a dependable source of financing for sustainable development projects in China, as well as promoting environmental sustainability and social fairness. Similar to this, (Lo, 2016; Zhang and Li, 2018) discovered that carbon markets can offer a market-based mechanism for encouraging emissions reduction and natural resource management practices in China. According to the current study's findings, green bonds and carbon markets are the best green financing options for implementing sustainable development and resource management techniques in China. Overall, the findings of this study are in line with those of earlier research that has highlighted the significance of adopting green financing mechanisms in fostering sustainable development and natural resource management practices in China. In order to attain SDGs, the study offers significant insights into the precise criteria and sub-criteria that are critical for evaluating and ranking green financing mechanisms for natural resource management in China.

5. Conclusion

The COVID-19 epidemic makes the findings of this research even more relevant. The evaluation and prioritizing of green financing options for natural resource management in China can help policymakers and stakeholders address the pandemic's economic and sustainability implications. This study therefore, applied a hybrid fuzzy MCDM method based on AHP and VIKOR to scrutinize and prioritize green financing mechanisms for natural resource management in China to accomplish the SDGs. Six criteria and eighteen sub-criteria have been identified in the research as essential for evaluating the effectiveness of green financing strategies. Six green financing strategies were evaluated: carbon markets, green bonds, green investment funds, public-private partnerships, microfinance, and sustainable tourism financing. The study has revealed insights into financial sustainability, institutional feasibility, and environmental impact. These factors have been identified as significant in encouraging sustainable development and the proper management of natural resources in China. The results further stress the role of carbon markets, green bonds, and green investment funds as the foremost green financing options for advancing sustainable development and effective resource management practices in China. Overall, there have been three essential contributions made in this study. First, this research contributes to a more nuanced understanding of green financing mechanisms and their role in promoting sustainable development and effective natural resource management. It provides a detailed and analytical lens through which the interconnections and implications of different green financing strategies can be understood, evaluated, and applied in China's efforts to meet the SDGs. Second, from the model perspective, this study contributes to using a unique hybrid fuzzy MCDM method, combining AHP and VIKOR, to evaluate green financing mechanisms for natural resource management in China. This approach overcomes the limitations of standalone models, enhancing decision-making efficiency in the often uncertain environmental policy field. This innovative model integrates multiple factors as financial sustainability, institutional feasibility, and environmental impact, which is a rarity in the existing literature. Third, from the literature perspective, this research fills the gap by examining green financing mechanisms' role in China's sustainable development and natural resource management. This study expands the green financing and sustainable development literature and offers key insights into these mechanisms in China, one of the world's most rapidly developing economies, guiding policy decisions and future research.

The current study proposes policy implications for green financing mechanisms in China, specifically green bonds and carbon markets, to achieve sustainable development goals. It suggests that policymakers should prioritize institutional feasibility and establish institutional structures and regulatory frameworks to support green finance initiatives. It further suggests that financial sustainability, institutional feasibility, and environmental impact can be used as criteria to guide the design and assessment of green financing mechanisms. For policymakers, it is essential to take a holistic approach to sustainable development by fostering partnerships with NGOs, private sector organizations, and other stakeholders to promote green financing mechanisms. This should include consideration of social justice and environmental sustainability and the use of state-of-the-art technology in green finance initiatives. It is also equally important to take some initiatives to raise awareness of the potential of green finance is essential for public support. China can promote sustainable development and climate change mitigation by leveraging green finance mechanisms. These include green bonds, carbon markets, and green development banks, which facilitate investments in renewable energy and other environmentally friendly projects. Policy frameworks should support these mechanisms, focusing on transparency, risk management, and environmental standards. Regular monitoring and evaluation are also necessary to ensure the mechanisms' effectiveness, and research should be invested in exploring new and existing solutions further.

The study has several shortcomings that could be addressed in follow-up studies. First, the weights of the criteria and sub-criteria were determined using expert judgments and subjective assessments. Although fuzzy MCDM techniques frequently incorporate expert judgments, they can have biases and limits. For the purpose of validating the findings and offering a more objective evaluation, future study may take into account incorporating additional techniques, such as open surveys or data-driven methodologies. Second, the study only analyzed six green finance strategies, which may not have been sufficient to cover all potential avenues for achieving sustainable growth and effective resource management in China. In order to provide a more thorough analysis of the most effective strategies for achieving SDGs in China. Finally, this study was solely concerned with China's natural resource management, it may not be applicable to other countries. Future studies could expand the study to other regions to provide a more thorough analysis of green finance strategies for attaining SDGs and best practices for managing the world's natural resources.

CRediT author statement

Yijia Dai: Conceptualization, Methodology, Software. Xuanyuan Chen: Data curation, Writing- Original draft preparation. Xuanyuan Chen: Visualization, Investigation. Yijia Dai: Supervision. Xuanyuan Chen: Software, Validation. Yijia Dai: Writing- Reviewing and Editing. Xuanyuan Chen: Writing- Reviewing and Editing.

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Declaration of competing interest

It is confirmed that the authors have followed all ethical guidelines and that this work has not been submitted or is not being considered for publication elsewhere. The authors declare that there is no conflict of interest with anyone else.

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